

## 1 部分正确性注释程序

### 定义 1.1 (部分正确性注释程序)

对断言  $P, Q$ , 程序  $S$  和某字符串  $S^*$ , 递归定义

$$\{P\} S^* \{Q\} \vdash_{\text{PO}} \{P\} S \{Q\},$$

称  $\{P\} S^* \{Q\}$  是  $\{P\} S \{Q\}$  的**部分正确性注释程序**,  $\{P\}, \{Q\}$  分别为前后断言 (pre, post),  $S$  为骨架程序 (sk):

1. 当  $P \rightarrow Q$  时,  $\{P\} \text{skip} \{Q\} \vdash_{\text{PO}} \{P\} \text{skip} \{Q\}$ ,
2. 当  $P \rightarrow Q[u := t]$  时,  $\{P\} u := t \{Q\} \vdash_{\text{PO}} \{P\} u := t \{Q\}$ ,
3. 如果  $\{P\} S_1^* \{R\} \vdash_{\text{PO}} \{P\} S_1 \{R\}$  且  $\{R\} S_2^* \{Q\} \vdash_{\text{PO}} \{R\} S_2 \{Q\}$ , 则
 
$$\{P\} S_1^* \{R\} S_2^* \{Q\} \vdash_{\text{PO}} \{P\} S_1; S_2 \{Q\},$$
4. 如果  $\{P \wedge B\} S_1^* \{Q\} \vdash_{\text{PO}} \{P \wedge B\} S_1 \{Q\}$  且  $\{P \wedge \neg B\} S_2^* \{Q\} \vdash_{\text{PO}} \{P \wedge \neg B\} S_2 \{Q\}$ , 则
 
$$\begin{aligned} & \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\} \\ & \vdash_{\text{PO}} \{P\} \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ fi } \{Q\}, \end{aligned}$$
5. 如果  $P \rightarrow I$ ,  $\{I \wedge B\} S^* \{I\} \vdash_{\text{PO}} \{I \wedge B\} S \{I\}$  且  $I \wedge \neg B \rightarrow Q$ , 则
 
$$\begin{aligned} & \{P\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\} \\ & \vdash_{\text{PO}} \{P\} \text{while } B \text{ do } S \text{ od } \{Q\}. \end{aligned}$$

### 定理 1.1 (正规形定理)

对任意的断言  $P, Q$  和程序  $S$ ,  $\vdash_{\text{PW}} \{P\} S \{Q\}$  当且仅当存在  $\{P\} S^* \{Q\} \vdash_{\text{PO}} \{P\} S \{Q\}$ .

### 定义 1.2 (部分正确性注释程序上的操作)

把  $\{P\} E \{P\}$  也视为自身的注释程序, 递归定义注释程序带状态的一步转移关系:

1.  $\langle \{P\} \text{skip} \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{Q\} E \{Q\}, \sigma \rangle$ ,
2.  $\langle \{P\} u := t \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{Q\} E \{Q\}, \sigma[u := \sigma(t)] \rangle$ ,
3. 任给注释程序  $\{P\} S_1^* \{R\}$  和  $\{R\} S_2^* \{Q\}$ , 如果  $\langle \{P\} S_1^* \{R\}, \sigma \rangle \rightarrow \langle \{P'\} S_1'^* \{R\}, \sigma' \rangle$ , 那么
 
$$\langle \{P\} S_1^* \{R\} S_2^* \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{P'\} S_1'^* \{R\} S_2^* \{Q\}, \sigma' \rangle,$$
4. 任给注释程序  $\{P\} S_1^* \{R\}$  和  $\{R\} S_2^* \{Q\}$ , 如果  $\langle \{P\} S_1^* \{R\}, \sigma \rangle \rightarrow \langle \{R\} E \{R\}, \sigma' \rangle$ , 那么
 
$$\langle \{P\} S_1^* \{R\} S_2^* \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{R\} S_2^* \{Q\}, \sigma' \rangle,$$
5. 任给注释程序  $\{P \wedge B\} S_1^* \{Q\}$  和  $\{P \wedge \neg B\} S_2^* \{Q\}$ , 如果  $\sigma \models B$ , 则
 
$$\langle \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{P \wedge B\} S_1^* \{Q\}, \sigma \rangle,$$
6. 任给注释程序  $\{P \wedge B\} S_1^* \{Q\}$  和  $\{P \wedge \neg B\} S_2^* \{Q\}$ , 如果  $\sigma \not\models B$ , 则
 
$$\langle \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{P \wedge \neg B\} S_2^* \{Q\}, \sigma \rangle,$$
7. 任给  $P \rightarrow I$ , 注释程序  $\{I \wedge B\} S^* \{I\}$  和  $I \wedge \neg B \rightarrow Q$ , 如果  $\sigma \models B$ , 则
 
$$\begin{aligned} & \langle \{P\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \\ & \langle \{I \wedge B\} S^* \{I\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}, \sigma \rangle, \end{aligned}$$
8. 任给  $P \rightarrow I$ , 注释程序  $\{I \wedge B\} S^* \{I\}$  和  $I \wedge \neg B \rightarrow Q$ , 如果  $\sigma \not\models B$ , 则
 
$$\langle \{P\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{ od } \{Q\}, \sigma \rangle \xrightarrow{\text{def}} \langle \{Q\} E \{Q\}, \sigma \rangle.$$

## 定理 1.2

对任意的  $\langle G, \sigma \rangle \rightarrow \langle G', \sigma' \rangle$ ,

1.  $\text{post}(G') = \text{post}(G)$ ,
2. 如果  $\text{sk}(G') = E$ , 那么  $G$  一定形如  $\{P\} E \{P\}$ ,
3.  $\langle \text{sk}(G), \sigma \rangle \rightarrow \langle \text{sk}(G'), \sigma' \rangle$ ,
4. 如果  $\sigma \models \text{pre}(G)$ , 那么  $\sigma' \models \text{pre}(G')$ .

证明.

1. 2. 显然. 3. 4. 对转移操作归纳证明.

i)  $P \rightarrow Q$  且  $G = \{P\} \text{skip} \{Q\}$ , 则  $G' = \{Q\} E \{Q\}$ ,  $\sigma' = \sigma$ . 显然

$$\langle \text{skip}, \sigma \rangle \rightarrow \langle E, \sigma \rangle,$$

并且如果  $\sigma \models P$ , 那么  $\sigma \models Q$ .

ii) 有  $Q_1$  使得  $Q_1[u := t]$  是自由替换,  $P \rightarrow Q_1[u := t]$  且  $Q_1 \rightarrow Q$ ,  $G = \{P\} u := t \{Q\}$ , 则

$$G' = \{Q\} E \{Q\}, \quad \sigma' = \sigma[u := \sigma(t)],$$

显然  $\langle u := t, \sigma \rangle \rightarrow \langle E, \sigma[u := \sigma(t)] \rangle$ , 并且如果  $\sigma \models P$ , 那么  $\sigma \models Q_1[u := t]$ , 从而  $\sigma' = \sigma[u := t] \models Q_1$ , 即  $\sigma' \models Q$ .

iii)  $G = \{P\} S_1^* \{R\} S_2^* \{Q\}$ , 假设  $\langle \{P\} S_1^* \{R\}, \sigma \rangle \rightarrow \langle \{P'\} S_1'^* \{R\}, \sigma' \rangle$  并且

$$\langle S_1, \sigma \rangle \rightarrow \langle S_1', \sigma' \rangle, \quad \sigma \models P \rightarrow \sigma' \models P',$$

那么  $G' = \{P'\} S_1'^* \{R\} S_2^* \{Q\}$ , 此时  $\langle S_1; S_2, \sigma \rangle \rightarrow \langle S_1'; S_2, \sigma' \rangle$ .

iv)  $G = \{P\} S_1^* \{R\} S_2^* \{Q\}$ , 假设  $\langle \{P\} S_1^* \{R\}, \sigma \rangle \rightarrow \langle \{R\} E \{R\}, \sigma' \rangle$  并且

$$\langle S_1, \sigma \rangle \rightarrow \langle E, \sigma' \rangle, \quad \sigma \models P \rightarrow \sigma' \models R,$$

那么  $G' = \{R\} S_2^* \{Q\}$ , 此时  $\langle S_1; S_2, \sigma \rangle \rightarrow \langle S_2, \sigma' \rangle$ .

v)  $G = \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\}$  且  $\sigma \models B$ , 则

$$G' = \{P \wedge B\} S_1^* \{Q\}, \quad \sigma' = \sigma,$$

显然  $\langle \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ fi}, \sigma \rangle \rightarrow \langle S_1, \sigma \rangle$ , 并且如果  $\sigma \models P$ , 那么  $\sigma' = \sigma \models P \wedge B$ .

vi)  $G = \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\}$  且  $\sigma \not\models B$ , 则

$$G' = \{P \wedge \neg B\} S_2^* \{Q\}, \quad \sigma' = \sigma,$$

显然  $\langle \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ fi}, \sigma \rangle \rightarrow \langle S_2, \sigma \rangle$ , 并且如果  $\sigma \models P$ , 那么  $\sigma' = \sigma \models P \wedge \neg B$ .

vii)  $P \rightarrow I$ ,  $I \wedge \neg B \rightarrow Q$ ,  $G = \{P\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{od } \{Q\}$  且  $\sigma \models B$ , 则

$$G' = \{I \wedge B\} S^* \{I\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{od } \{Q\}, \quad \sigma' = \sigma,$$

显然

$$\langle \text{while } B \text{ do } S \text{ od}, \sigma \rangle \rightarrow \langle S; \text{while } B \text{ do } S \text{ od}, \sigma \rangle,$$

并且如果  $\sigma \models P$ , 那么  $\sigma \models I$ , 从而  $\sigma' = \sigma \models I \wedge B$ .

viii)  $P \rightarrow I$ ,  $I \wedge \neg B \rightarrow Q$ ,  $G = \{P\} \{\text{inv} : I\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{od } \{Q\}$  且  $\sigma \not\models B$ , 则

$$G' = \{Q\} E \{Q\}, \quad \sigma' = \sigma,$$

显然

$$\langle \text{while } B \text{ do } S \text{ od}, \sigma \rangle \rightarrow \langle E, \sigma \rangle,$$

并且如果  $\sigma \models P$ , 那么  $\sigma \models I$ , 从而  $\sigma' = \sigma \models I \wedge \neg B$ , 即  $\sigma' \models Q$ . □

## 2 完全正确性注释程序

### 定义 2.1 (完全正确性注释程序)

对断言  $P, Q$ , 程序  $S$  和某字符串  $S^*$ , 递归定义

$$\{P\} S^* \{Q\} \vdash_{\text{TO}} [P] S [Q],$$

称  $\{P\} S^* \{Q\}$  是  $[P] S [Q]$  的**完全正确性注释程序**, 前后断言与骨架程序的记法 pre, post, sk 同上节:

1. 当  $P \rightarrow Q$  时,  $\{P\} \text{skip} \{Q\} \vdash_{\text{TO}} [P] \text{skip} [Q]$ ,
2. 当  $P \rightarrow Q[u := t]$  时,  $\{P\} u := t \{Q\} \vdash_{\text{TO}} \{P\} u := t \{Q\}$ ,
3. 如果  $\{P\} S_1^* \{R\} \vdash_{\text{TO}} [P] S_1 [R]$  且  $\{R\} S_2^* \{Q\} \vdash_{\text{TO}} [R] S_2 [Q]$ , 则

$$\{P\} S_1^* \{R\} S_2^* \{Q\} \vdash_{\text{TO}} [P] S_1; S_2 [Q],$$

4. 如果  $\{P \wedge B\} S_1^* \{Q\} \vdash_{\text{TO}} [P \wedge B] S_1 [Q]$  且  $\{P \wedge \neg B\} S_2^* \{Q\} \vdash_{\text{TO}} [P \wedge \neg B] S_2 [Q]$ , 则

$$\begin{aligned} & \{P\} \text{if } B \text{ then } \{P \wedge B\} S_1^* \{Q\} \text{ else } \{P \wedge \neg B\} S_2^* \{Q\} \text{ fi } \{Q\} \\ & \vdash_{\text{TO}} [P] \text{if } B \text{ then } S_1 \text{ else } S_2 \text{ fi } [Q], \end{aligned}$$

5. 如果  $P \rightarrow I$ ,  $\{I \wedge B\} S^* \{I\} \vdash_{\text{TO}} [I \wedge B] S [I]$ ,  $t$  是整型项,  $z$  是整型变量,  $z$  不出现在  $t, B, I, S$  中,  $\{I \wedge B \wedge t = z\} S^{**} \{t < z\} \vdash_{\text{TO}} [I \wedge B \wedge t = z] S [t < z]$ ,  $I \rightarrow t \geq 0$  且  $I \wedge \neg B \rightarrow Q$ , 则

$$\begin{aligned} & \{P\} \{\text{inv} : I\} \{\text{bd} : t\} \text{while } B \text{ do } \{I \wedge B\} S^* \{I\} \text{od } \{Q\} \\ & \vdash_{\text{TO}} [P] \text{while } B \text{ do } S \text{od } [Q]. \end{aligned}$$

### 定理 2.1 (正规形定理)

对任意的断言  $P, Q$  和程序  $S$ ,  $\vdash_{\text{TW}} [P] S [Q]$  当且仅当存在  $\{P\} S^* \{Q\} \vdash_{\text{TO}} [P] S [Q]$ .

完全正确性注释程序上的操作几乎与定义 1.1 相同 (只是多了 **bd** 标记), 定理 1.2 的四条性质对完全正确性注释程序也依旧成立.